

Task_6_Battery_Management_Sy

WOKWI

SAVE

SHARE

Simulation

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sketch.ino

diagram.json

libraries.txt

Library Manager

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Protection Status: RECOVERY WAIT

Relay State: OFF

Buzzer State: OFF

Relay Feedback: LOW

WiFi RSSI: 8

Blynk Status: CONNECTED

Last Fault Log: [10040 ms] SENSOR FAULT

00:13.695

15%

RECOVERY WAIT
Relay OFF Safe

Devices - Blynk.Console

blf1.blynk.cloud/dashboard/715168/global/devic...

My organization - 7025M

Messages used 183 of 100.0k

Battery Intelligence System

Live

Pack Voltage

13.15V

0 15

RSSI

0dBm

-100 0

Voltage Trend

16:10 16:15 16:20 16:25

Health Status

HEALTHY

Operator Reco...

Battery op

Runtime mode

FAULT HISTORY

[10040 ms]

Risk Level

LOW

Fault Status

RECOVER

Task_6_Battery_Management_Sy

WOKWI

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SHARE

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Library Manager

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Relay State: OFF

Buzzer State: ON

Relay Feedback: LOW

WiFi RSSI: 16

Blynk Status: CONNECTED

Last Fault Log: [17046 ms] CRITICAL IMBAL

Cloud Telemetry Sent to Blynk

00:18.128

12%

MODE: FAILSAFE
CRIT IMBAL

Devices - Blynk.Console

blf1.blynk.cloud/dashboard/715168/global/devic...

My organization - 7025M

Messages used 183 of 100.0k

Battery Intelligence System

Live

Pack Voltage

12.81V

0 15

RSSI

0dBm

-100 0

Voltage Trend

16:10 16:15 16:20 16:25

Health Status

CRITICAL

Operator Reco...

Balance b...

Runtime mode

FAILSAFE

[17046 ms]

Risk Level

HIGH

Fault Status

CRIT IMBA

Task_6_Battery_Management_Sy

WOKWI

SAVE

SHARE

sketch.ino

diagram.json

libraries.txt

Library Manager

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Simulation

00:26.828

15%

RECOVERY WAIT
Relay OFF Safe

Protection Status: NORMAL
Relay State: ON
Buzzer State: OFF
Relay Feedback: LOW
WiFi RSSI: 14
Blynk Status: CONNECTED
Last Fault Log: [25547 ms] SYSTEM RECOVERED

Devices - Blynk.Console

blf1.blynk.cloud/dashboard/715168/global/devic...

Battery Intelligence System

Health Status

HEALTHY

Operator Reco...

Battery op

Runtime mode

FAULT HISTORY

[19535 ms]

Risk Level

LOW

Fault Status

NORMAL

Pack Voltage

13.15V

0

15

RSSI

0dBm

-100

0

Voltage Trend

16:30

16:35

16:40

16:45

16:50

16:55

17:00

16:30

16:35

16:40

16:45

16:50

16:55

17:00

Region: EU/US

Privacy Policy

Terms of Service

Devices - Blynk.Console

blf1.blynk.cloud/dashboard/715168/global/devic...

Battery Intelligence System

Health Status

PACK FAILURE

Operator Recommendation

Replace weak b...

Runtime mode

SHUTDOWN

Fault History

[16429 ms] WEA

Fault Status

WEAK CELL

Risk Level

CRITICAL

Pack Voltage

11.83V

0

15

Battery SOC

57%

0

100

RSSI

0dBm

-100

0

Battery Cell 1

3.3V

0

4

Battery Cell 2

3.3V

0

4

Battery Cell 3

1.93V

0

4

Battery Cell 4

3.3V

0

4

Voltage Trend

17:00

17:05

17:10

17:15

17:20

17:25

17:30

17:00

17:05

17:10

17:15

17:20

17:25

17:30

Region: EU/US

Privacy Policy

Terms of Service

Task_6_EXTRA - Wokwi ESP

WOKWI

SAVE

SHARE

sketch.ino

diagram.json

libraries.txt

Library Manager

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Simulation

00:19.793

24%

MODE=SHUTDOWN
NEW_CELL

Protection Status: WEAK CELL
Relay State: OFF
Buzzer State: ON
Relay Feedback: LOW
WiFi RSSI: 13
Blynk Status: CONNECTED
Last Fault Log: [19354 ms] WEAK CELL

Blynk Console | **Ask Gemini**

blr1.blynk.cloud/dashboard/715168/global/devic...

Battery Intelligence System

Pack Voltage: 13.2V | Battery SOC: 53% | RSSI: 0dBm

Battery Cell 1: 3.3V | Battery Cell 2: 3.3V | Battery Cell 3: 1.8V | Battery Cell 4: 3.3V

Health Status: **HEALTHY** | Runtime Status: **NORMAL** | Fault Status: **RECOVERY WAIT**

Operator Recommendation: **Replace weak ba** | Fault History: **[22428 ms] WEA** | Risk Level: **CRITICAL**

Voltage Trend

WOKWI | **Task_6_EXTRA - Wokwi ESP32** | **Ask Gemini**

wokwi.com/projects/467800632500734977

sketch.ino

```
libraries.txt
Library Manager
213 void loop() {
246   int (currenttu
249   }
250   }
251   }
252   float convertAd
253   return adcVal
254   }
255   }
256   void updatePre
257   for (int i =
258     battery.pre
259   }
260   }
261   }
262   void readCellVo
263   for (int i =
264     int adcValu
265     battery.cel
266   }
267   }
268   }
269   void calculateP
270   battery.packV
271   for (int i =
272     battery.pac
```

Simulation

00:26.193 19%

Protection Status: RECOVERY WAIT
Relay State: OFF
Buzzer State: OFF
Relay Feedback: LOW
WiFi RSSI: 14
Blynk Status: CONNECTED
Last Fault Log: [23350 ms] RAPID CHANGE

Blynk Console | **Ask Gemini**

blr1.blynk.cloud/dashboard/715168/global/devic...

Battery Intelligence System

Pack Voltage: 13.2V | Battery SOC: 99% | RSSI: 0dBm

Battery Cell 1: 3.3V | Battery Cell 2: 3.3V | Battery Cell 3: 3.3V | Battery Cell 4: 3.3V

Health Status: **HEALTHY** | Runtime Status: **NORMAL** | Fault Status: **NORMAL**

Operator Recommendation: **Battery operating** | Fault History: **[29354 ms] SYS** | Risk Level: **LOW**

Voltage Trend

WOKWI | **Task_6_EXTRA - Wokwi ESP32** | **Ask Gemini**

wokwi.com/projects/467800632500734977

sketch.ino

```
libraries.txt
Library Manager
213 void loop() {
246   int (currenttu
249   }
250   }
251   }
252   float convertAd
253   return adcVal
254   }
255   }
256   void updatePre
257   for (int i =
258     battery.pre
259   }
260   }
261   }
262   void readCellVo
263   for (int i =
264     int adcValu
265     battery.cel
266   }
267   }
268   }
269   void calculateP
270   battery.packV
271   for (int i =
272     battery.pac
```

Simulation

00:32.426 16%

Protection Status: NORMAL
Relay State: ON
Buzzer State: OFF
Relay Feedback: LOW
WiFi RSSI: 6
Blynk Status: CONNECTED
Last Fault Log: [29354 ms] SYSTEM RECOVERED

Devices - BlynkConsole

blr1.blynk.cloud/dashboard/715168/global/devic...

BlynkConsole

My organization - 70258

Battery Intelligence System - Online

My organization - 70258

Dashboard

Custom Data

Developer Zone

Devices

Automations

Users

Organizations

Locations

Snapshots

Panel Management

In-App Messaging

Predictive Maintenance

Pack Voltage

9.96V

015

Battery SOC

99%

0100

RSSI

0dBm

-1000

Battery Cell 1

3.3V

04

Battery Cell 2

3.3V

04

Battery Cell 3

3.3V

04

Battery Cell 4

3.3V

04

Health Status

PACK FAILURE

Runtime mode

SHUTDOWN

WiFi Status

WEAK CELL

Operator Recommendation

Battery operating

Fault History

[33428 ms] RELAY

Fault Level

LOW

Voltage Trend

7:0013:00:0013:00:0013:00:0013:00:0013:00:00

WOKWI

Task_6_EXTRA - Wokwi ESP8266

SAVE

SHARE

sketch.ino

diagram.json

libraries.txt

Library Manager

213void loop() {

214// (current

215}

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Simulation

00:38.193

19%

MODE: SHUTDOWN

WEAK CELL

Protection Status: WEAK CELL

Relay State: OFF

Buzzer State: ON

Relay Feedback: LOW

WiFi RSSI: 7

Blynk Status: CONNECTED

Last Fault Log: [37356 ms] WEAK CELL

19:40

26-06-2026